GMI German-Malaysian Institute

DIPLOMA PROGRAMME

ASSIGNMENT

Academic Period: JULY 2026

Program	:	**CYBERSECURITY**
Course	:	**DIGITAL FORENSICS AND INCIDENT RESPONSE**
Code	:	**CBS 2363**
Total Marks	:	**100**

GENERAL INSTRUCTIONS:

1. Write down clearly (with capital letters) your name, trade, group and the date of examination at the bottom of this page.
2. Do not open this question booklet until you are told to do so by the invigilator.
3. Read the specific instructions and questions carefully before answering.

This paper consists of **4** printed pages, including this page.

Student's Name	:	**ABDUL HARIZ MOHD SARNUDIN**
Student's I/D No.	:	**CBS24070208**
Semester/Trade/Group	:	**SEM 5 / CBS / DCBS 1**
Date	:	**20/08/2026**

No	Assessment Criteria	CLO	PO	Level of Diff	Full Marks	Score	Marks	Comment
1.	Demonstration	1	2	C3	30			
2.	Interpreting Findings (Analysis)	1	2	C3	30			
3.	Fluency & Clarity	1	2	C2	15			
4.	QnA	1	2	C2	15			
5.	Report Content & Format	1	2	C1	10			
				Total	100			
				Total Marks	/100		/30	
Total			**PO 2**		/30			
				Signature				

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Scenario

You are required by your superior to demonstrate a data **recovery tool** available on the Internet. The tool should be available as an open-source software which can be demonstrated in Linux/Windows environments.

1. **Introduction:**
    - Provide an overview of data recovery and its importance.
2. **Tool Preparation:**
    - Determine the website to download the tool.
    - Perform tool installation and configuration.
3. **Step-by-Step Installation:**
    - Screen Recording:
        i. Clearly show each step of the service installation process.
        ii. Make sure no important steps are skipped.
        iii. Use any screen recording tool.
        iv. Ensure the wallpaper in the video is your face to avoid plagiarism.
        - Video duration should be between 10-15 minutes ***(depending on the complexity of the installation).***
4. **Submission**
    - Submission in the form of video screen recording.

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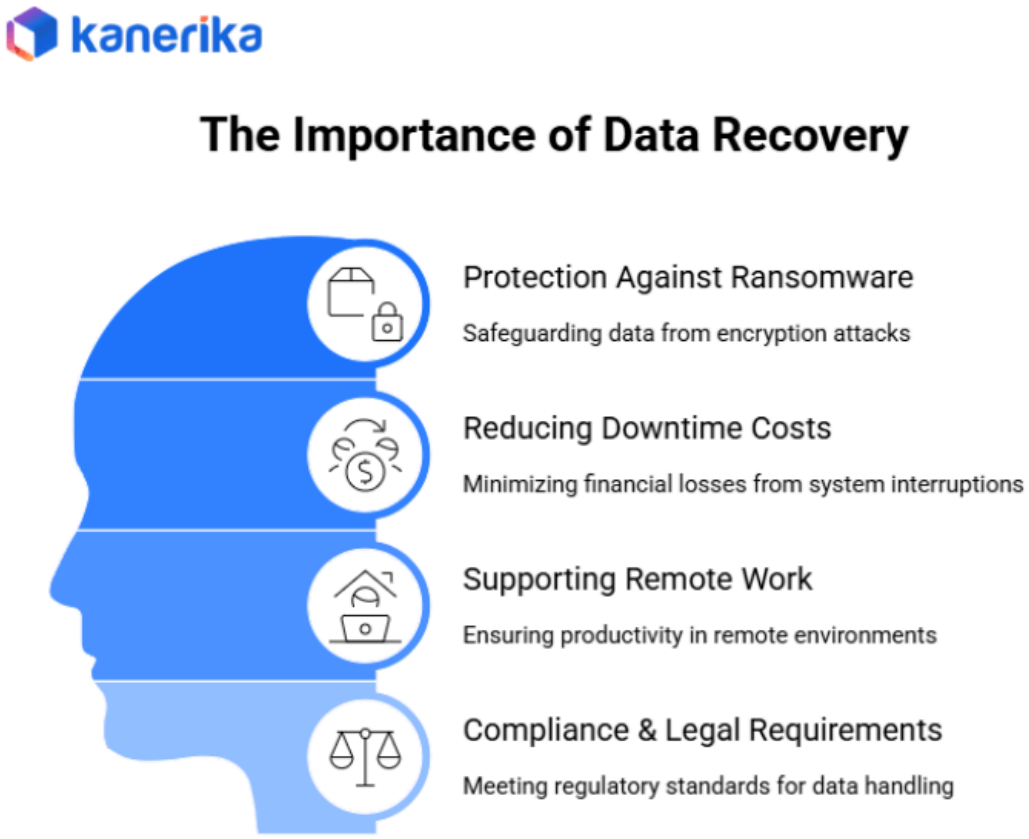
NO	CRITERIA	APPROACHES EXPECTATIONS M[1]	MEETS EXPECTATIONS M[2]	EXCELLENT M[3]	WEIGHT	MARKS
ASSIGNMENT – 15%						
1	**Demonstration of tools**	Able to slightly demonstrate tool used	Able to demonstrate tool used	Able to demonstrate tool with full details	M/3*30	
2	**Interpreting Findings** (Analysis/concept knowledge)	Able to slightly interpret the findings and results	Able to interpret most of the findings and results	Able to interpret all of the findings and results	M/3*30	
3	**Fluency and Clarity** Presentation	Not fluent and voice is not clear.	Quite fluent and voice is slightly clear to be heard	Very fluent and voice is easy to be heard	M/3*15	
4	**QnA**	Not able to answer questions, not prepared and no confidence at all	Able to answer questions but with little preparation and confidence	Able to all answer questions very well and confidently. Very well prepared	M/3*15	
5	**Report Content & Format**	Report is poorly presented (unclear and confusing)	The report is clearly presented, but not enough.	The report is clearly presented and in details	M/3*10	
	TOTAL					

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Introduction

Data recovery is a crucial process aimed at retrieving lost, corrupted, or inaccessible data from storage devices. Whether data loss occurs due to accidental deletion, formatting, hardware failure, software corruption, virus infection or physical damage to the storage device, data recovery specialists employ a series of steps to recover and restore the lost data.

Forensic data recovery often forms part of a Digital Forensics and Incident Response (DFIR) process. While incident response focuses on containing and eradicating threats, forensic data recovery helps preserve and recover digital evidence to support investigations, compliance activities and potential legal proceedings.



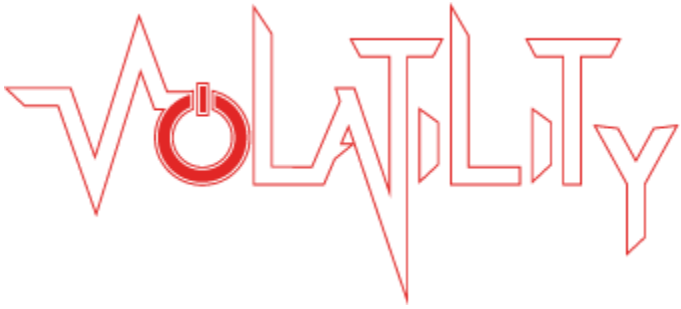
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Tool Preparation

In this lab we are using Volatility 3, an open-source memory forensics framework maintained by the Volatility Foundation for analyzing volatile RAM samples. We will run Volatility 3 on Ubuntu 24.04 Desktop on VMware Workstation Pro.

Website & Download Source

- **Primary Repository:** GitHub - volatilityfoundation/volatility3
- **Official Foundation Website:** Volatility Foundation



A little bit about Volatility. First released in 2007, The Volatility Framework was developed as an open source memory forensics tool written in Python. It has remained free and available to the world, and it is actively maintained by members of The Volatility Project. Volatility development is supported by The Volatility Foundation.

Volatility releases are the result of significant in-depth research into OS internals, applications, malicious code, and suspect activities. Releases represent a milestone in not only our team's progress, but also in the development of the community and forensics capabilities as a whole. While releases may seem few and far between, we strive to perform rigorous testing of our new features before calling it stable.

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Tool Installation & Configuration

OS : Ubuntu 24.04 Desktop
Environment : VMware Workstation Pro





Sources :

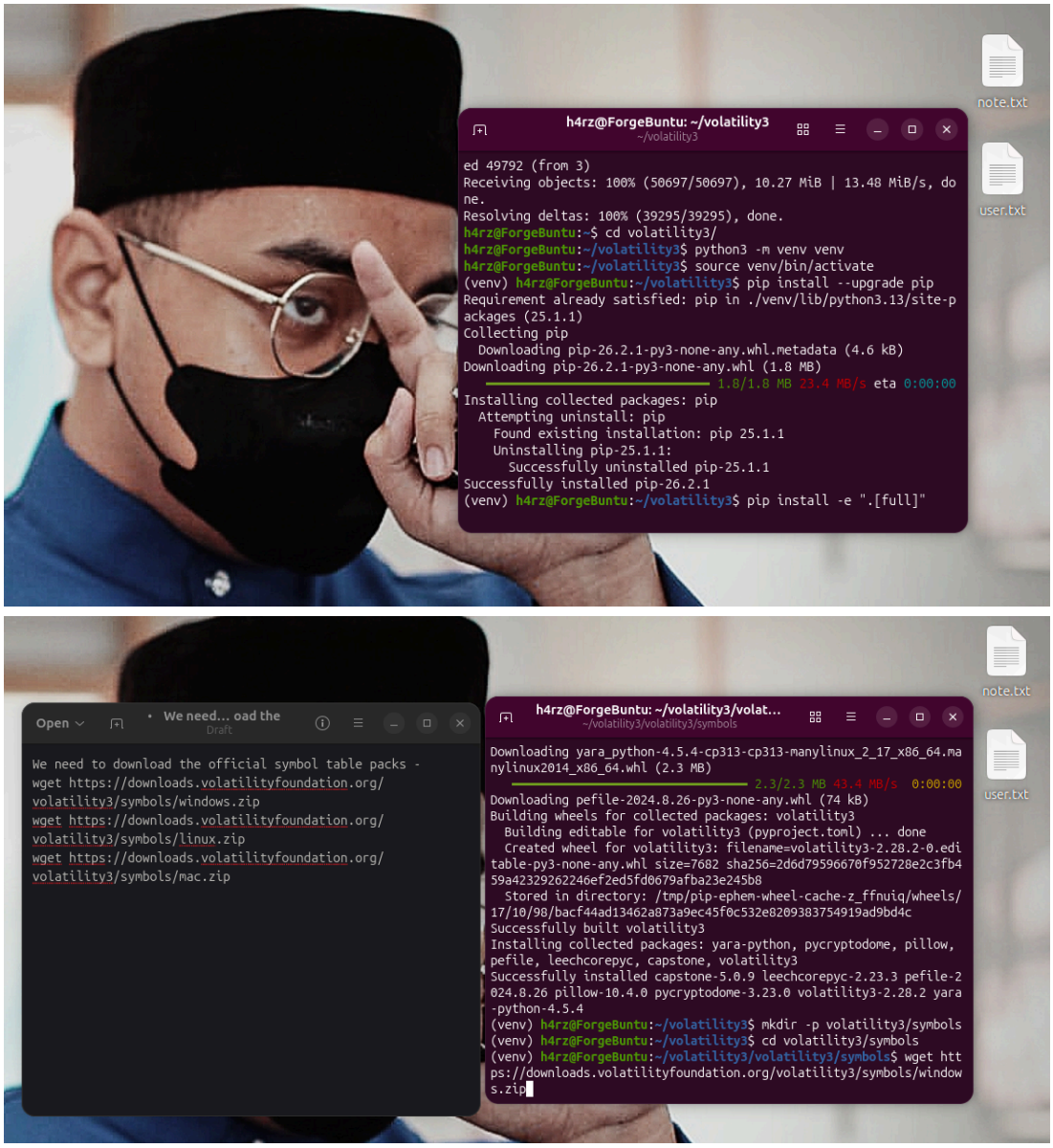
- https://ubuntu.com/download/desktop
- https://www.vmware.com/products/desktop-hypervisor/workstation-and-fusion

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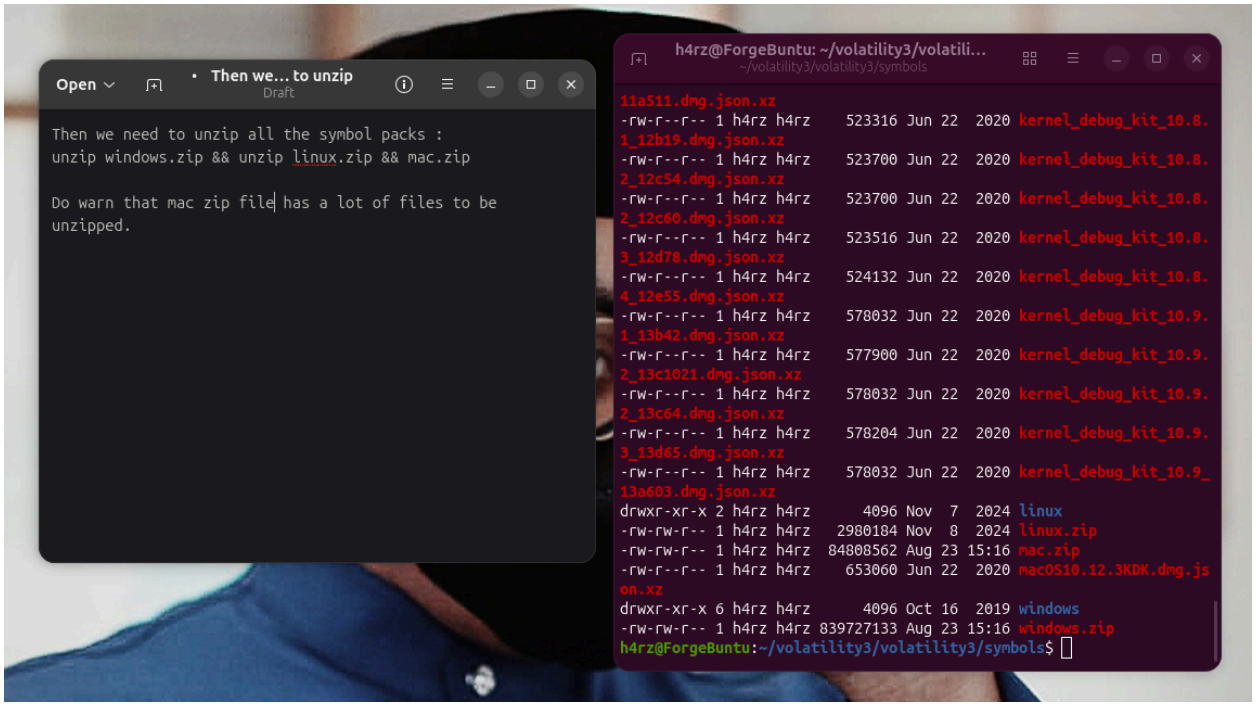


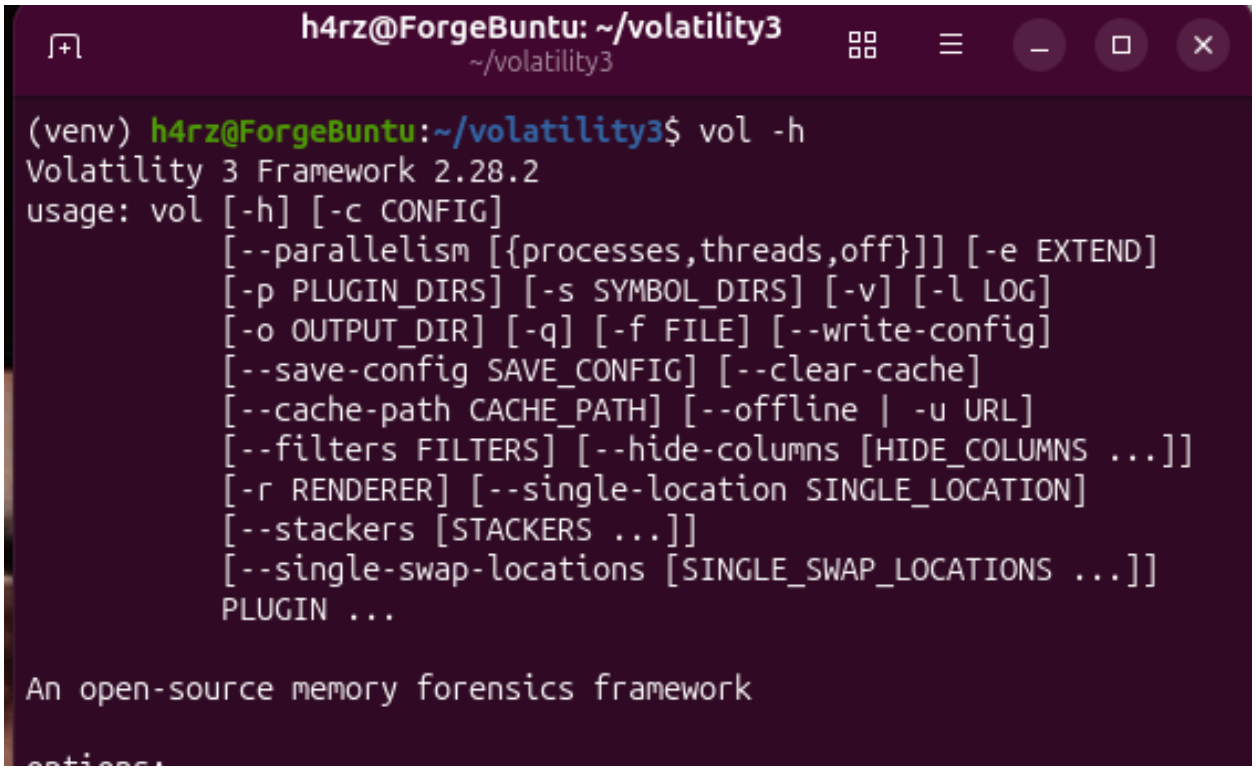
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Volatility 3 uses intermediate symbol format (ISF) tables instead of legacy profiles.

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Verify installation by using "vol -h", and it will display "An open-source memory forensics framework" that determines the installation completion without errors.

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Video Installation & Demonstration

Overview

This video provides a step-by-step walkthrough demonstrating the installation of Volatility 3, environment configuration, and execution of core memory forensic plugins against the target memory sample.

Attribute	Details
Title	Volatility 3 Installation & Forensic Demonstration
Platform	Streamable
Direct URL	https://streamable.com/pfbdql
Duration	05:36
Resolution	720p

Watch Here

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Summary

In this lab, we focused on strengthening our digital forensics and data recovery capabilities by learning how to effectively navigate, configure and troubleshoot different investigative tools. Rather than relying on a single automated framework, We worked through system setup challenges and dependency issues to better understand how forensic utilities interact with volatile and persistent data.

This hands-on experience deepened our ability to extract critical evidence ranging from volatile runtime traces to hidden filesystem artifacts while building the technical adaptability required to overcome unexpected software barriers. Ultimately, exploring a wider variety of digital forensics tools expanded our problem-solving toolkit, ensuring that we can remain resourceful and adapt to complex scenarios during real-world investigations.

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References

- GeeksforGeeks. (2024, February 19). *What is data recovery?* https://www.geeksforgeeks.org/techtips/what-is-data-recovery/

- 4Secure. (n.d.). *Forensic data recovery*. Retrieved February 23, 2026, from https://www.4-secure.com/forensic-data-recovery/

- Tomer, S., Apurva, A., Ranakoti, P., Yadav, S., & Roy, N. R. (2017, October). *Data recovery in forensics*. 2017 International Conference on Computing and Communication Technologies for Smart Agriculture (IC3SA), Gurgaon, India. ResearchGate. https://www.researchgate.net/publication/323058776_Data_recovery_in_Forensics

- The Volatility Foundation. (n.d.). *The Volatility Framework*. Retrieved February 23, 2026, from https://volatilityfoundation.org/the-volatility-framework/

- Volatility Foundation. (n.d.). *Volatility 3 documentation*. Read the Docs. Retrieved February 23, 2026, from https://volatility3.readthedocs.io/en/stable/

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